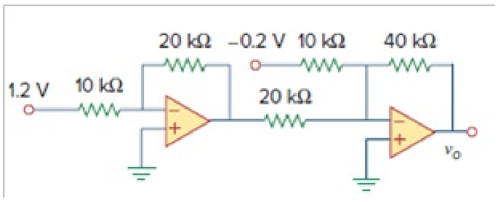
## Chapter 5, Problem 61P

### Problem

Determine *vo* in the circuit of Fig. 5.88.

**Figure. 5.88:**




### Step-by-step solution

**Step 1** of 3

Refer to Figure 5.88 in the textbook and assign nodes, the modified circuit is shown in Figure 1.

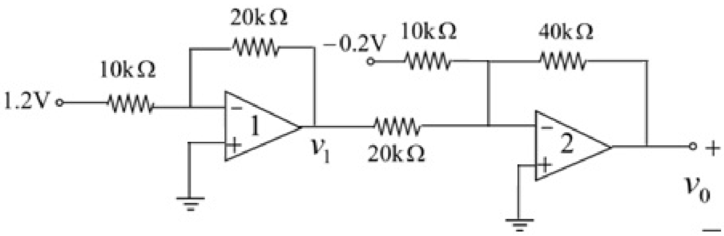



Figure 1

**Step 2** of 3

In Figure 1, the first op-amp circuit is act as an inverting amplifier circuit. The output voltage is,

$$
\begin{aligned}
v_1 &= -\left(\frac{R_f}{R_i}\right)v_i \\
&= -\left(\frac{20\times10^3}{10\times10^3}\right)(1.2) \\
&= -2(1.2) \\
v_1 &= -2.4\,\text{V}
\end{aligned}
$$

**Step 3** of 3

In Figure 1, the second op-amp circuit is act as a two input summing amplifier circuit. The output voltage is,

$$\begin{aligned} v_0 &= -R_f\left(\frac{v_1}{R_1}+\frac{v_2}{R_2}\right) \\ &= -40\times10^3\left(\frac{-0.2}{10\times10^3}+\frac{v_1}{20\times10^3}\right) \\ &= 0.8-2v_1 \end{aligned}$$

Substitute $-2.4\,\mathrm{V}$ for $v_1$.

$$\begin{aligned} v_0 &= 0.8-2(-2.4) \\ &= 0.8+4.8 \\ v_0 &= 5.6\,\mathrm{V} \end{aligned}$$

Therefore, the output voltage, $v_0$ is $\boxed{5.6\,\mathrm{V}}$.